Introduction

The NBA markets itself as a league of parity, but underneath the upsets and storylines, there are patterns—beauty in chaos. Certain teams consistently overperform in the playoffs, while others fall short of expectations time and time again. Analysts, front offices, and fans alike wonder: *which team statistics matter most, and which ones just look good on paper?*

The league has undergone a transformation over the past decade into an entirely new product, a metamorphosis kick-started by one organization—one man, in fact. Eleven years ago, Stephen Curry and the Golden State Warriors permanently changed the geometry of the NBA. Teams now prioritize spacing, pace, and shooting in a way that would have been deemed reckless in the 1990s. It is widely assumed that three-point efficiency is the currency of winning basketball in the modern era—but assumptions without data do not a theorem make.

To formally test which statistics best predict playoff outcomes, this project uses eleven seasons of NBA data (2014-15 through 2024-25), covering all 30 franchises. For each of the 330 individual team seasons, we constructed a dependent variable that encodes playoff success on a 0–5 scale: 0 for teams that missed the playoffs entirely, 1 for those that lost in the first round, 2 for those that lost in the second, 3 for the third, 4 for a loss in the finals, and 5 for the NBA's champion. We then assembled a set of independent variables that reflect three layers of information:

1. Traditional box-score statistics such as three-point percentage (3P%), offensive and defensive rebounding, steals, blocks, and turnover rate.
2. Advanced efficiency measures such as True Shooting Percentage (TS%) and Pace.
3. A global team performance metric, Net Rating, is defined as point differential per 100 possessions.

We progress in stages, first testing whether three-point percentage truly is the most important traditional stat for predicting playoff success. Next, we examine whether a more comprehensive offensive efficiency metric (TS%) supersedes raw 3P%. Finally, we ask whether Net Rating, which bundles both sides of the ball into a single number, dominates both. Using multiple regression with standardized variables, our *coup de grace* will be the derivation of a composite "Playoff Predictor Index" that combines the most informative stats into one interpretable measure.

The goal is not to produce an initially high R^2—which is the measure of the percentage of change in our dependent variable that our work predicts—but to hone our model, and in doing so tell a coherent story about how modern NBA teams win in the spring: whether Curry truly turned the league on its head or a more all-encompassing metric, an evergreen truth, remains the best predictor of playoff success.

Statement of Hypotheses

This project is organized around three explicit hypotheses that mirror how a typical NBA fan or analyst might reason through the problem, starting from a simplistic narrative and moving toward a more holistic understanding.

Hypothesis 1: Three-Point Percentage

The first hypothesis reflects the modern era's emergent wisdom:

Among traditional counting and percentage statistics, a team's three-point percentage (3P%) is the strongest positive predictor of playoff success.

The rationale is straightforward. Modern offenses are built around spacing and three-point shooting. Teams that shoot well from deep force defenses to stretch, opening driving lanes and stressing rotations. Intuitively, a team that can consistently hit a high volume of threes should have an inherent advantage in a

seven-game series, where variance tends to average out. If any single box-score stat "should" predict playoff performance in today's NBA, it's 3P%.

Hypothesis 2: True Shooting Percentage

The second hypothesis takes a more nuanced view of scoring efficiency:

Once introduced into the model, True Shooting Percentage (TS%) will outperform three-point percentage as a predictor of playoff success.

TS% incorporates two-point field goals made, three-point field goals made, and free throws into a single efficiency metric. From a modeling perspective, TS% should dominate 3P% because it captures the entirety of a team's scoring profile, proverbially swallowing up the latter's impact. A team that lives at the rim and the line can be just as effective as a perimeter-centric offense, and TS% is designed to reflect that.

Hypothesis 3: Net Rating

The third hypothesis widens the scope beyond offense:

Net Rating will outperform all of our offensive efficiency metrics and become our new playoff prediction champion.

Net Rating effectively encodes how much a team outscored (or was outscored by) its opponents. It is not just an offensive stat; it reflects overall dominance—teams that consistently outscore opponents by larger margins tend to be both efficient offensively and dominant defensively. If the playoffs are ultimately about how much better you are than your opponent across all possessions, Net Rating should emerge as our model's cleanest predictor of playoff success.

Data Collection Process

All data collected was from publicly available pages on Basketball Reference, a standard source for NBA analytics. For each team-season, we noted our dependent variable's measurement: how each team performed in the playoffs, if they made it. Our initial independent variables included field goal percentage (FG%), 3-pointers made, 3PT%, 2-pointers made, 2P%, free throws made, FT%, offensive rebounds, defensive rebounds, assists, steals, blocks, and turnovers. We added more over time: TS%, offensive rebound percentage, turnover percentage, Pace, and eventually Net Rating.

To make coefficients directly comparable across variables, we standardized each predictor into a z-score, producing adjusted versions such as ADJ3P, ADJDRB, ADJSTL, and so on. These standardized variables have a mean of 0 and a standard deviation of 1. There is no sampling design in the traditional sense, as the data represent a population of NBA team-seasons in the post-Curry modern era.

Data Description

The dependent variable, playoff round reached (Playoff_Rd), is an ordered discrete variable taking values from 0 to 5. As expected, its distribution is heavily skewed toward the lower end. In any given season, only one team can win the championship and be awarded a 5, only one can be awarded a 4, two can be awarded a 3, four teams can get a 2, and eight teams can get a 1. By contrast, 14 teams fail to make the playoffs every year, earning themselves a resounding 0. Across the 330 team-seasons, therefore, a large mass of teams have values 0 or 1, representing non-playoff teams and first-round exits, and skewing the distribution accordingly (Exhibit 1).

The skew is informative: most teams do not experience deep playoff success, which makes it statistically harder to predict those rare outcomes. Any model that explains a meaningful amount of variation in this

variable is performing well in a noisy environment dominated by injuries, matchups, and year-to-year randomness.

The predictive variables show ranges consistent with modern NBA play:

- **Three-Point Percentage (3P%)** ranges from the low 30s to around 40% over the years. Teams at the top end are often elite offenses or heavily spacing-oriented lineups.
- **True Shooting Percentage (TS%)** is higher on average than 3P%, as it incorporates all scoring opportunities. Across the sample, TS% exhibits variation, but hovers around the 50% mark.
- **Net Rating (NetRtg)** spans from strongly negative (roughly –10 points per 100 possessions) for bottom-tier teams to strongly positive (roughly +10) for title contenders. The distribution is roughly centered near zero, as expected in a balanced league where some teams win as often as others lose.
- Other variables, such as ORB%, DRB, TOV%, STL, BLK, and Pace, exhibit more modest spreads. For example, Pace varies across eras and coaching philosophies—some teams push the tempo, others deliberately slow the game down—but its variation is smaller in magnitude than the variation in Net Rating.

Standardizing these predictors into adjusted z-scores (e.g., ADJTS, ADJNET) allows a direct interpretation: a one-unit change in an adjusted predictor corresponds to a one-standard-deviation change in the original stat.

To understand how these variables relate to one another, we computed a correlation matrix among the predictors (Exhibit 2). Several patterns emerge:

- TS% and 3P% are moderately correlated, with a correlation of about 0.69. This implies that roughly 48% of the variance in one can be explained by the other ($0.69^2 \approx 0.48$). That is substantial overlap but not perfect redundancy.

- 3P% and Net Rating are also moderately correlated (around 0.61), and TS% and Net Rating exhibit a similar correlation (around 0.58). Strong shooting and overall point differential tend to travel together; elite teams are rarely inefficient on offense.
- Correlations between the remaining variables and our premier trio are generally smaller in magnitude, often below 0.5 in absolute value.

From a multicollinearity perspective, these correlations suggest noticeable but not catastrophic overlap. In particular, TS% and 3P% are closely related measures of shooting efficiency, so it shouldn't surprise that 3P% loses unique explanatory power once TS% enters the model. Similarly, Net Rating envelops offensive efficiency, so its introduction naturally absorbs much of their predictive signal.

The key point is that the correlations remain well below the usual "red flag" threshold ($|r| > 0.8$). As a result, while multicollinearity influences coefficient magnitudes and significance levels, it does not render the regression estimates unusable or unstable.

Scatterplots between each predictor and the corresponding playoff round support our regression assumptions:

- The relationship between 3P% and playoff success is positive but noisy (Exhibit 3). Many middling shooting teams make deep runs, and some elite shooting teams exit early.
- TS% vs. playoff success shows a slightly stronger positive pattern, consistent with TS% being a more comprehensive measure of scoring quality (Exhibit 4).
- Net rating vs. playoff success demonstrates the clearest linear relationship. Teams with Net Ratings one or more standard deviations above the mean tend to reach later rounds, while teams with negative Net Ratings rarely advance (Exhibit 5).

Methodology

Given the continuous, ordered nature of the dependent variable and the compounded linear relationships observed in the scatterplots, we used multiple linear regression as the primary modeling framework, with an emphasis on interpretability.

To test the three hypotheses, we ran a sequence of multiple nested regression models using our standardized predictors:

- **Model 1 (Basic Stats Model):**

  Playoff_Rd = f(ADJ3P, ADJORB%, ADJDRB, ADJTOV%, ADJSTL, ADJBLK)

- **Model 2 (True Shooting and Pace Model):**

  Playoff_Rd = Model 1 + ADJTS + ADJPACE

- **Model 3 (Net Rating Model):**

  Playoff_Rd = Model 2 + ADJNET

Model performance is evaluated primarily using the coefficient of determination, R^2, and Adjusted R^2, which penalizes the addition of extra predictors. Statistical significance is assessed via p-values for each coefficient. Because the models are nested, changes in R^2 and Adjusted R^2 as we move from Model 1 → Model 2 → Model 3 provide a clear measure of how much each new block of variables improves explanatory power.

Using the coefficients from the final model, we constructed a "Playoff Predictor Index" as a weighted linear combination of the standardized predictors. This index is then ranked within each season and mapped back to predicted playoff rounds, allowing a straightforward evaluation of how well the model predicts team success.

Results

Model 1 (Exhibit 6)**:**

Confirming our first hypothesis, among basic stats, 3PT% emerges as the most impactful, with the highest coefficient and the lowest p-value among those included. Turnover rate also shows up as an important factor in the opposite direction: higher turnover rates are associated with reduced playoff success, as intuition would suggest. Other variables—offensive rebounding, defensive rebounding, steals, and blocks—contribute, but their standardized coefficients are smaller in magnitude and less statistically robust.

The model explains a meaningful but imperfect share of the variance in playoff outcomes. The measured R^2 was a moderate 0.33, suggesting that while traditional stats, especially 3P% and TOV%, do matter, they only account for about a third of the variance in playoff performance on their own.

Model 2 (Exhibit 7)**:**

As soon as True Shooting joins the equation, the model shifts the way we predicted. It steps forward as the superior indicator, absorbing much of the explanatory power that previously sat with 3P%. TS% has a standardized coefficient seven times larger than that of 3P%, and its p-value makes it unlikely to be due to random variation. This is exactly what we expected, given the moderate correlation between the two front-runners (around 0.69); TS% is a more robust encapsulation of shooting efficiency.

Pace has an unexpected—perhaps surprising—impact on our model; with the second largest coefficient after TS% and a slightly lower p-value, it proves to be a detrimental factor for playoff success, and even more deterministic than 3PT%. For every one-standard-deviation increase in pace, according to model 2, teams can expect to reach 0.61 fewer playoff rounds. This might be due to experience—younger teams play faster, while older teams are slower and methodical—but Pace's pronounced negative impact should not be ignored.

Crucially, the addition of TS% and Pace significantly improves model fit. Both R^2 and Adjusted R^2 increase relative to Model 1, with R^2 reaching 0.47, indicating that these variables bring genuine explanatory power rather than just inflating R^2 through overfitting. These results continue to support our hypotheses: TS% outperforms 3P% as a predictor of playoff success because it reflects a broader view of scoring efficiency.

Model 3 (Exhibit 8)**:**

Our final model introduces Net Rating to our existing stats. Its impact, as expected, is dramatic, becoming by far the most pertinent predictor of playoff success. The standardized coefficient on ADJNET is close to 0.8, meaning that a one-standard-deviation increase in Net Rating predicts roughly 0.8 additional playoff rounds on average, holding all other variables constant. Its p-value is also extremely small, affirming the strength and reliability of the relationship.

Once Net Rating is included, the relative importance of the other predictors shifts. TS% remains somewhat useful, but its coefficient shrinks significantly; 3P% matters very little on its own at this stage, as both TS% and Net Rating now carry overlapping information with it. Turnovers and Pace become similarly impactful to one another, but in opposite directions. Pace continues to hurt performance, but about six times less than in model 2, while turnovers seem to become beneficial. Both of their p-values are high enough, however, to completely dismiss these findings as noise, which explains that counterintuitive conclusion. Rebounding and defense-related stats still contribute around the margins, but Net Rating dominates them too.

The overall fit of the model improves substantially. The final R^2 is approximately 0.52, meaning that just over half of the variance in playoff success across these 330 team-seasons is explained by our third and final model. The Adjusted R^2 is just above 0.50, indicating that this improvement is not merely an artifact

of adding more predictors. For an outcome as noisy and context-dependent as playoff performance, explaining about half the variance with regular-season team stats is strong evidence that the model is capturing something real—not just fitting noise.

The Playoff Predictor Index

Using the coefficients from the final standardized regression, we constructed an algorithm (Exhibit 9) to predict a team's playoff round reached, and compared it to actual measured success. In our evaluation, we ranked teams within each season based on their indexed value (1 through 30) and then mapped those ranks to predicted playoff rounds (Exhibit 10). The model achieves over 50% accuracy when predicting the exact playoff round and performs even better when grouping teams into broader tiers (e.g., non-playoff, early exit, deep run, champion). That level of performance makes the index a useful summary statistic for a team's playoff potential.

Recommendations and Conclusion

The results of this analysis have clear implications for how we think and talk about NBA team quality. The data validates some of the modern conventional wisdom: three-point shooting does matter. In a model restricted to traditional box-score style stats, 3P% is the strongest positive predictor of playoff success. Teams that cannot shoot from deep are swimming upstream against the math.

The results also demonstrate the limitations of focusing on any single offensive stat. Once True Shooting is introduced, because it reflects the full scoring profile—inside scoring, three-point shooting, and free throws—it proves superior to 3PT%. This underscores an often forgotten aspect of the game among locker room talk and barber shop chop; Steph Curry did not "ruin" basketball. What matters most now is the same as what has always mattered: smart scoring reigns supreme.

Net Rating being added to the regression expectedly strengthened our predictive power, reflecting the nuance with which NBA discourse should be broached. It becomes the dominant variable by a wide margin, with a one-standard-deviation increase corresponding to nearly an extra playoff round.

From a practical standpoint, if we were advising a front office or an analytics-minded coaching staff, we would recommend that they track Net Rating obsessively, use TS% as their primary efficiency metric (not just 3PT%), and treat other box-score stats as context. The peripherals matter, but personnel differences between teams likely explain their variance.

This project reinforces a simple, data-backed truth about the modern NBA: style matters, but substance—consistently outscoring your opponents over the course of a season—is what wins in the playoffs. Teams that generate strong Net Ratings are the ones that tend to still be playing in June.

Appendices

Exhibit 1 (Playoff Round Histogram):

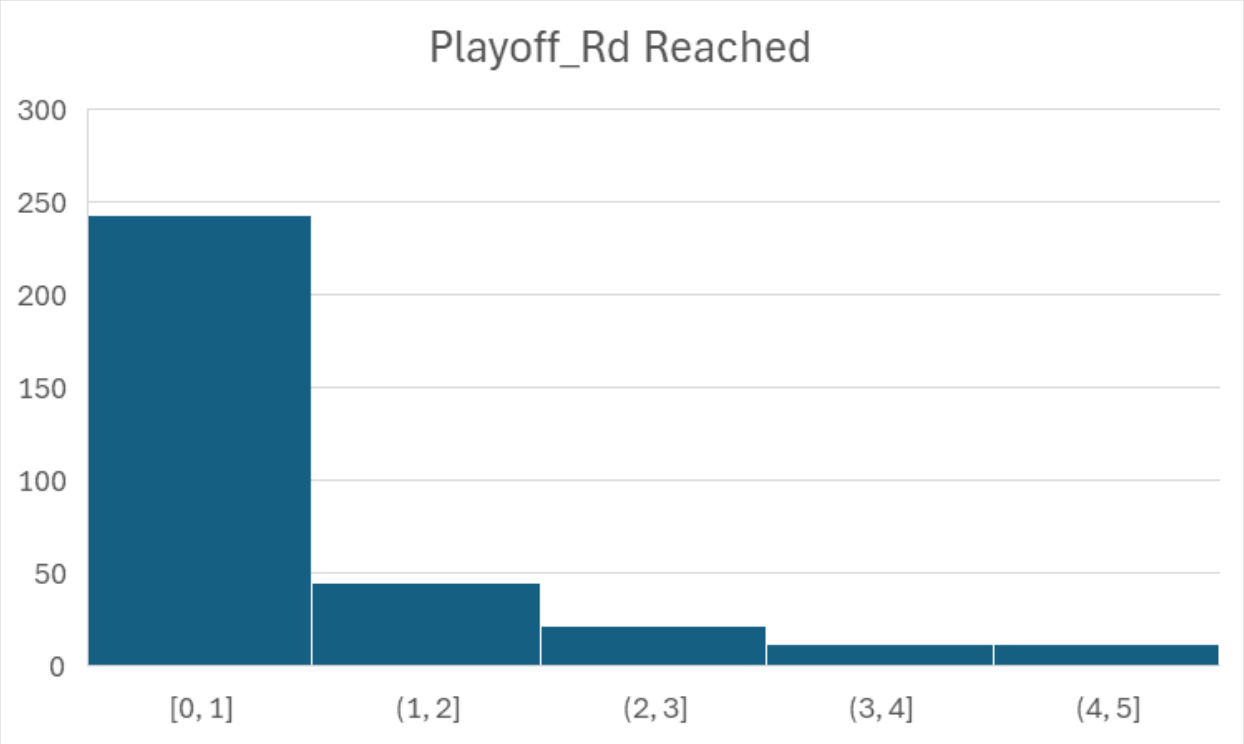


Exhibit 2 (Correlation Matrix):

	TS%	*3P%*	*ORB%*	*DRB*	*TOV%*	*STL*	*BLK*	*Pace*	*NetRtg*
TS%	1								
3P%	0.693808	1							
ORB%	-0.12868	-0.15518	1						
DRB	0.2849	0.269962	-0.17183	1					
TOV%	-0.25943	-0.22608	0.060275	-0.13177	1				
STL	-0.0172	-0.05785	0.126336	-0.26072	0.123328	1			
BLK	0.15081	0.114636	0.009649	0.225635	-0.01045	0.19402	1		
Pace	0.414256	0.060253	-0.15974	0.443179	-0.03597	0.121482	0.161621	1	
NetRtg	0.57822	0.609025	0.112368	0.433936	-0.34235	0.169084	0.283386	-0.0616	1

Exhibit 3 (3PT% vs. Playoff_Rd):

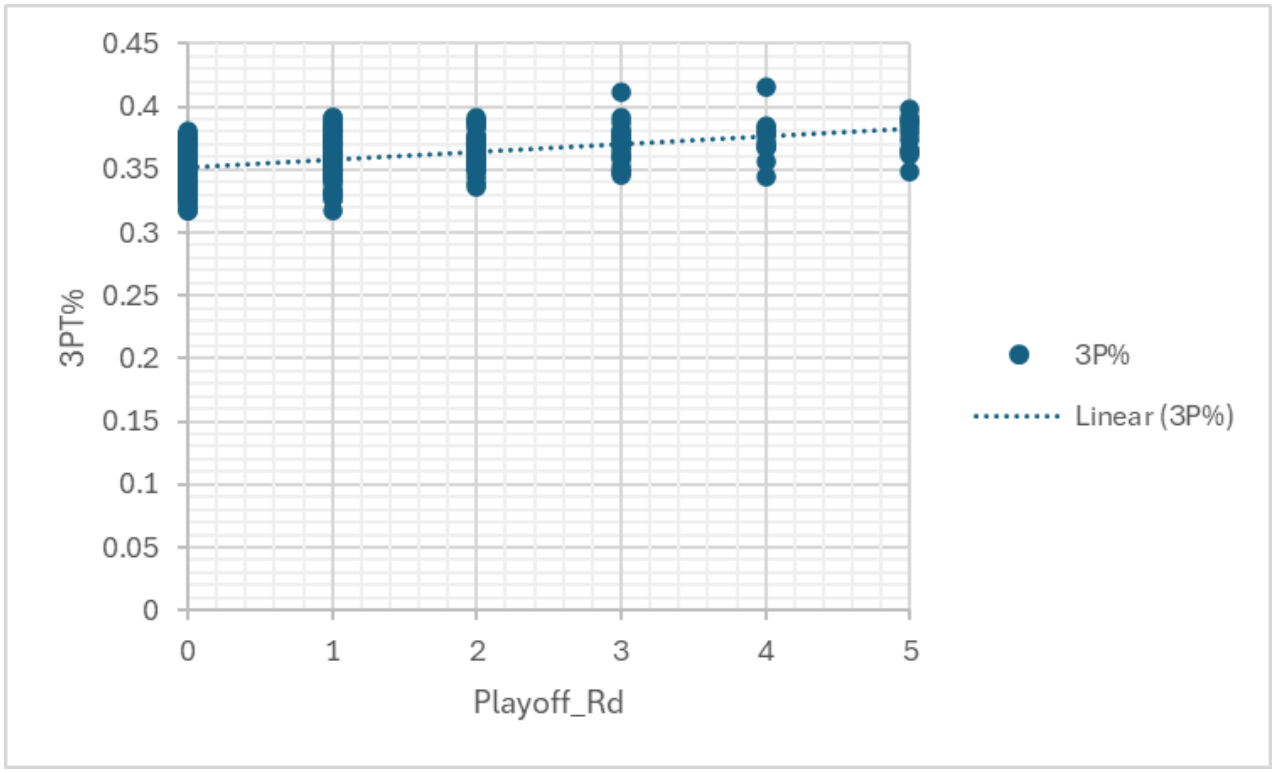


Exhibit 4 (TS% vs. Playoff_Rd):

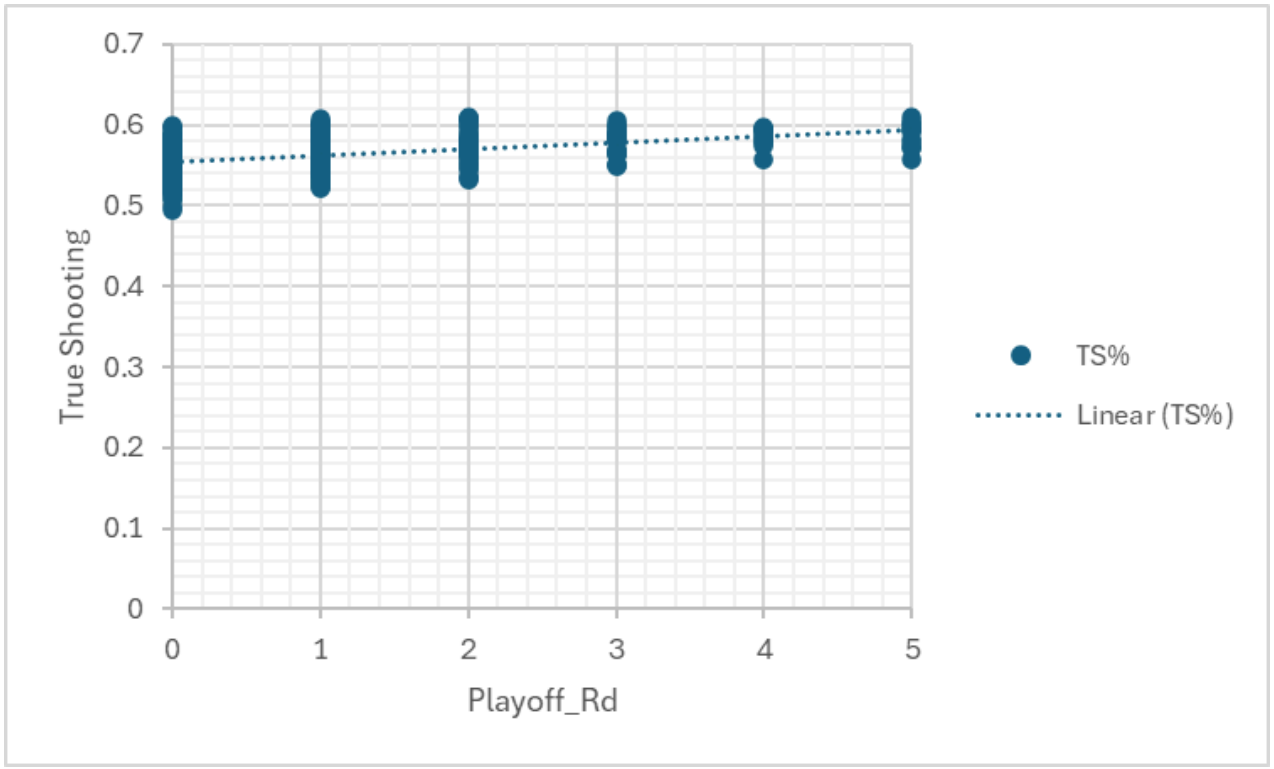


Exhibit 5 (Net Rating vs. Playoff_Rd):

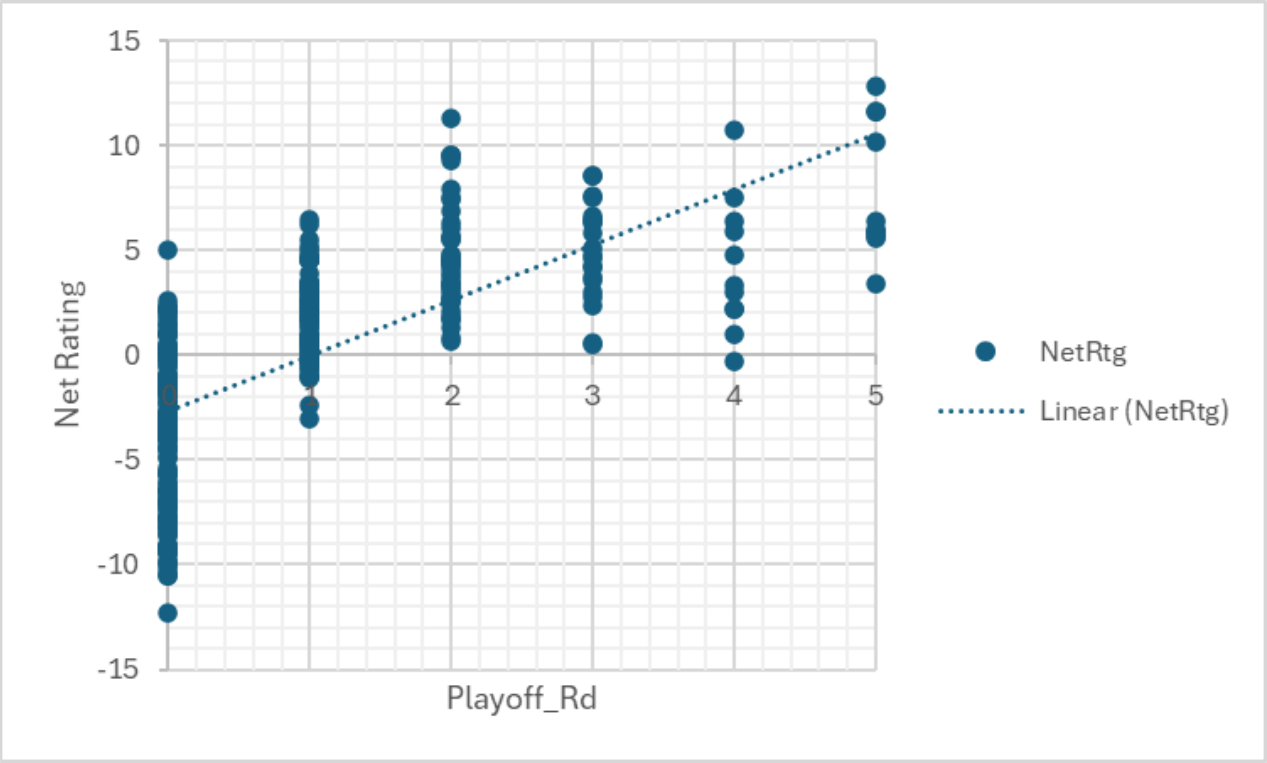


Exhibit 6 (Model 1):

Regression Statistics	
Multiple R	0.575923747
R Square	0.331688163
Adjusted R Square	0.319273701
Standard Error	1.078295366
Observations	330

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	6	186.3926656	31.06544427	26.71788592	8.12274E-26
Residual	323	375.5588495	1.162720896		
Total	329	561.9515152			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-21.89807003	2.28408655	-9.587233038	2.54398E-19	-26.39163483	-17.40450523	-26.39163483	-17.40450523
3P%	34.74987864	3.871786921	8.97515265	2.37E-17	27.13277437	42.36698292	27.13277437	42.36698292
ORB%	0.070834312	0.024863965	2.848874362	0.004668905	0.021918549	0.119750075	0.021918549	0.119750075
DRB	0.192108421	0.040630736	4.728155086	3.38928E-06	0.112174127	0.272042714	0.112174127	0.272042714
TOV%	-4.268572285	6.377693799	-0.66929715	0.503784059	-16.81563639	8.278491818	-16.81563639	8.278491818
STL	0.305997972	0.078960431	3.875333119	0.000128962	0.150656305	0.461339639	0.150656305	0.461339639
BLK	0.108768047	0.088206153	1.233111784	0.218430727	-0.064763059	0.282299152	-0.064763059	0.282299152

Exhibit 7 (Model 2):

Regression Statistics	
Multiple R	0.684594177
R Square	0.468669187
Adjusted R Square	0.455427298
Standard Error	0.964449313
Observations	330

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	8	263.36936	32.92117	35.39292414	6.26977E-40
Residual	321	298.5821552	0.930162477		
Total	329	561.9515152			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	1.012121212	0.053091187	19.06382703	1.01958E-54	0.907670581	1.116571843	0.907670581	1.116571843
ADJTS	0.662477334	0.088740584	7.46532536	7.88674E-13	0.487890732	0.837063936	0.487890732	0.837063936
ADJ3P	0.095893981	0.082745022	1.158909367	0.247354527	-0.066897062	0.258685024	-0.066897062	0.258685024
ADJORB	0.110740421	0.055172658	2.007161249	0.045570509	0.002194743	0.2192861	0.002194743	0.2192861
ADJDRB	0.561928307	0.068618898	8.189118777	6.29269E-15	0.426928742	0.696927872	0.426928742	0.696927872
ADJTOV	0.021739111	0.055743603	0.389983963	0.696807042	-0.087929834	0.131408056	-0.087929834	0.131408056
ADJSTL	0.376862371	0.060178721	6.262385898	1.21595E-09	0.258467857	0.495256886	0.258467857	0.495256886
ADJBLK	0.052343157	0.056911551	0.919728183	0.358405441	-0.059623587	0.164309901	-0.059623587	0.164309901
ADJPACE	-0.609423274	0.073161183	-8.329871826	2.37981E-15	-0.753359247	-0.465487302	-0.753359247	-0.465487302

Exhibit 8 (Model 3):

Regression Statistics	
Multiple R	0.722816129
R Square	0.522463156
Adjusted R Square	0.509032432
Standard Error	0.915752002
Observations	330

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	9	293.598962	32.62210689	38.90059581	2.15247E-46
Residual	320	268.3525531	0.838601729		
Total	329	561.9515152			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	1.012121212	0.050410489	20.0775918	1.33715E-58	0.912943366	1.111299058	0.912943366	1.111299058
ADJTS	0.206266019	0.113461186	1.817943437	0.070007709	-0.016958083	0.429490121	-0.016958083	0.429490121
ADJ3P	0.029103157	0.079350689	0.366766281	0.71403576	-0.127011783	0.185218097	-0.127011783	0.185218097
ADJORB	-0.020695357	0.056776936	-0.36450289	0.715723267	-0.132398583	0.091007868	-0.132398583	0.091007868
ADJDRB	0.068678116	0.104853955	0.654988317	0.512945814	-0.137612078	0.274968309	-0.137612078	0.274968309
ADJTOV	0.161158871	0.057798824	2.788272473	0.005616029	0.047445177	0.274872566	0.047445177	0.274872566
ADJSTL	0.051497539	0.078751069	0.653928128	0.513627757	-0.103437706	0.206432785	-0.103437706	0.206432785
ADJBLK	0.000654667	0.054719432	0.011964076	0.990461731	-0.107000614	0.108309949	-0.107000614	0.108309949
ADJPACE	-0.116857749	0.107499957	-1.087049263	0.277833157	-0.328353698	0.094638201	-0.328353698	0.094638201
ADJNET	0.796384695	0.13264307	6.003967578	5.22151E-09	0.535422058	1.057347331	0.535422058	1.057347331

Exhibit 9 (Our Algorithm):

```
Adjusted Prediction = 1.012121212
      + 0.206266019 * ADJTS
      + 0.095893981 * ADJ3P
      - 0.020069357 * ADJORB
      + 0.086878116 * ADJDRB
      + 0.161588871 * ADJTOV
      + 0.051497539 * ADJSTL
      + 0.006564867 * ADJBLK
      - 0.116857749 * ADJPACE
      + 0.796384695 * ADJNET
```

Exhibit 10 (Playoff Predictor in Action):

Team	Season	Playoff_Rd	Final Predicted Playoff Round
Atlanta Hawks	25	0	0
Boston Celtics	25	2	3
Brooklyn Nets	25	0	0
Charlotte Hornets	25	0	0
Chicago Bulls	25	0	0
Cleveland Cavaliers	25	2	4
Dallas Mavericks	25	0	0
Denver Nuggets	25	2	2
Detroit Pistons	25	1	1
Golden State Warriors	25	2	1
Houston Rockets	25	1	1
Indiana Pacers	25	4	1
Los Angeles Clippers	25	1	3
Los Angeles Lakers	25	1	1
Memphis Grizzlies	25	1	2
Miami Heat	25	1	1
Milwaukee Bucks	25	1	2
Minnesota Timberwolves	25	3	2
New Orleans Pelicans	25	0	0
New York Knicks	25	3	1
Oklahoma City Thunder	25	5	5
Orlando Magic	25	1	0
Philadelphia 76ers	25	0	0
Phoenix Suns	25	0	0
Portland Trail Blazers	25	0	0
Sacramento Kings	25	0	1
San Antonio Spurs	25	0	0
Toronto Raptors	25	0	0
Utah Jazz	25	0	0
Washington Wizards	25	0	0